

Variational Derivation of Euler's Equation for Inviscid Flow

Source: Cambridge Mathematical Tripos, Variational Principles, Example Sheet 2 (2023), Question 9.

The mass density $\rho(t, \mathbf{x})$ and velocity field $\mathbf{v}(t, \mathbf{x})$ of a compressible fluid are constrained by conservation of mass to satisfy the continuity equation

$$\dot{\rho} + \nabla \cdot (\rho \mathbf{v}) = 0. \quad (1.1)$$

Given that the energy density of the fluid is $u(\rho)$, the action (for inviscid irrotational flow) is

$$S[\rho, \mathbf{v}, \phi] = \int dt \int d^3x \left\{ \frac{1}{2} \rho |\mathbf{v}|^2 - u(\rho) + \phi [\dot{\rho} + \nabla \cdot (\rho \mathbf{v})] \right\}, \quad (1.2)$$

where $\phi(t, \mathbf{x})$ is a Lagrange multiplier field imposing the continuity condition (1.1). Find the Euler--Lagrange equations for this action. Show that they imply $\mathbf{v} = \nabla \phi$ (so ϕ is the velocity potential). Given that the fluid pressure $P(t, \mathbf{x})$ satisfies

$$\nabla P = \rho \nabla h(t, \mathbf{x}), \quad h = u'(\rho), \quad (1.3)$$

deduce Euler's equation for inviscid irrotational flow:

$$\rho [\dot{\mathbf{v}} + (\mathbf{v} \cdot \nabla) \mathbf{v}] = -\nabla P. \quad (1.4)$$

Solution. First, we consider the equation of motion of $\mathbf{v}$:

$$\frac{\partial \mathcal{L}}{\partial v_i} = -\rho v_i + \phi(\partial_i \rho), \quad \frac{\partial \mathcal{L}}{\partial(\partial_i v_j)} = \rho \phi \delta_{ij} \implies \nabla \phi = \mathbf{v}. \quad (1.5)$$

Then, for the EoM with respect to ρ , we find

$$\frac{\partial \mathcal{L}}{\partial \rho} = -\frac{1}{2} |\mathbf{v}|^2 - u'(\rho), \quad \frac{\partial \mathcal{L}}{\partial \dot{\rho}} = \phi \implies \dot{\phi} = -\frac{1}{2} |\mathbf{v}|^2 - u'(\rho). \quad (1.6)$$

Finally, the constraint is

$$\dot{\rho} + \nabla \cdot (\rho \mathbf{v}) = 0. \quad (1.7)$$

Putting them together, one can easily find

$$\rho [\dot{\mathbf{v}} + (\mathbf{v} \cdot \nabla) \mathbf{v}] = -\nabla P. \quad (1.8)$$